

LADPORTAL PLC Ladder Logic Simulator Project Report

Portable Windows desktop application for PLC learning and simulation

Student / Developer	Bhushan Hiralal Bhole
Project name	LADPORTAL
Project identity	LADPORTAL by Automation Club desktop application
Version	1.0.0
Project type	Portable Windows desktop application
Domain	PLC programming, ladder logic, industrial automation learning
Primary purpose	Help students create, understand, and simulate PLC ladder logic without real hardware
Deliverable	Project work report for academic, portfolio, and interview discussion

This report presents LADPORTAL as a student automation project and portfolio application. The project is framed as a learning tool for PLC ladder logic, I/O mapping, simulation workflow, and desktop packaging. It is not described as a replacement for certified industrial PLC engineering software.

The supplied deliverable is a Windows portable executable. For this report, the executable package was inspected statically for metadata, version information, packaging type, and file integrity. The application should be functionally tested on a Windows system before final submission or demonstration.

Abstract

LADPORTAL is a PLC ladder logic learning and simulation application packaged as a portable Windows desktop program. The project gives students a software environment where they can understand ladder rungs, contacts, coils, input and output mapping, simulation state, project organization, and basic automation control logic without requiring a physical PLC.

The project is useful because many students know PLC theory but do not always have access to industrial hardware, licensed software, training panels, or plant networks. LADPORTAL reduces that gap by providing a controlled offline environment for practice, explanation, and portfolio demonstration.

Why I Chose This Project

I chose this project because PLC programming is one of the most practical skills in industrial automation. In real factories and warehouses, PLCs control conveyors, sensors, actuators, motors, interlocks, alarms, and machine sequences. A simulator helps me explain these concepts even when I do not have direct access to real industrial hardware.

- To connect PLC theory with a visual and practical programming workflow.
- To create an accessible tool for students who want to practice ladder logic.
- To show my automation interest through a working desktop application, not only written notes.
- To support Automation Club style demos, workshops, and beginner training.
- To build a project that can be discussed in PLC, maintenance, WCS, or industrial automation interviews.

Problem Statement

PLC programming is usually taught with real PLC hardware or vendor software such as Siemens TIA Portal, CODESYS, Rockwell Studio 5000, or other industrial tools. These tools are powerful, but they may be expensive, heavy, hardware-dependent, or too complex for a beginner. The problem is to provide a simpler learning environment where a student can build and simulate basic ladder logic before moving to real industrial platforms.

Problem area	Explanation	Project response
Limited hardware access	Students may not have PLCs, I/O modules, sensors, motors, or training panels.	LADPORTAL can be used as a software-only practice environment.
Complex industrial tools	Professional PLC suites have many options that can confuse beginners.	The project focuses on a simpler ladder learning workflow.
Need for visual understanding	Ladder logic is easier to learn when rung states and I/O changes are visible.	The simulator can show logic state, inputs, outputs, and execution flow.
Portfolio visibility	A student needs project evidence that demonstrates automation thinking.	The portable app shows a concrete software tool connected to PLC concepts.

Project Objectives

- Create a desktop application that represents PLC ladder logic in a beginner-friendly way.
- Allow users to organize PLC learning work as projects rather than only isolated examples.
- Provide controls for simulation start, stop, input changes, and output observation.
- Help users understand the relationship between contacts, coils, rungs, and scan cycles.
- Package the tool as a portable Windows executable so it can be shared and demonstrated easily.
- Build a project that supports Automation Club workshops and technical interview discussion.

What The Project Is

LADPORTAL is a learning-oriented PLC ladder logic simulator. It can be presented as a simplified PLC programming portal where a user creates or opens a project, defines input and output points, builds ladder logic, starts simulation, changes input states, and observes output behavior. It is intended for education and practice, not for direct control of real machines.

Area	Meaning in the project	User benefit
PLC project workspace	A place to organize logic, I/O names, and project settings.	Makes learning structured.
Ladder editor	Visual area for rungs, contacts, coils, and simple logic elements.	Helps users see PLC logic graphically.
I/O manager	Defines digital inputs and outputs used in the ladder program.	Connects symbols to machine-style signals.
Simulation mode	Runs the logic and updates states as inputs change.	Shows cause and effect without hardware.
Monitoring view	Displays active and inactive logic states.	Makes debugging easier.
Portable package	Windows EXE package with application files inside.	Easy to carry, share, and demonstrate.

Project Scope

The scope is educational PLC simulation. The project should show core ladder logic concepts clearly and reliably, while keeping industrial-risk functions outside the first version. It should not be used as a live PLC controller, plant SCADA, safety PLC, or certified commissioning tool.

- Included: ladder learning, basic I/O configuration, simulation, project organization, and desktop packaging.
- Included: explanation of PLC scan cycle, input image, logic solve, and output update.
- Excluded: direct machine control, safety control, fieldbus communication, real PLC download, and plant network access.
- Excluded: production recipe handling, live alarms, and certified industrial validation.

My Work And Contribution

Work area	Contribution	Why it is important
Concept design	Defined LADPORTAL as a student-friendly PLC learning portal.	Gives the project a clear educational identity.
Automation workflow	Connected project tree, ladder editor, I/O list, and simulation workflow.	Matches how real PLC work is organized.

Work area	Contribution	Why it is important
User experience	Focused on simple controls such as project actions, simulation on/off, and readable panels.	Makes the app easier for beginners.
PLC explanation	Built the project around ladder logic concepts such as contacts, coils, rungs, and scan cycle.	Shows technical understanding.
Packaging	Prepared a Windows portable 1.0.0 build for sharing and demonstration.	Turns the project into a usable deliverable.
Documentation	Prepared project report style material for academic and interview use.	Makes the project easier to present professionally.

Technology And Tools Used

Technology or concept	Use in project	Reason
Windows desktop packaging	The submitted file is a portable Windows executable.	Easy to open and demonstrate on Windows.
NSIS wrapper	The file is a Nullsoft Installer self-extracting archive.	Common method for portable app delivery.
Embedded payload archive	The package contains an embedded 7z payload.	Keeps application files compressed inside one EXE.
PLC ladder logic	Core project subject.	Most common entry point for industrial automation programming.
Digital I/O model	Represents inputs, outputs, and signal state.	Connects software logic to real machine thinking.
Simulation engine	Evaluates ladder behavior without a physical PLC.	Supports practice and demonstration.

System Architecture

The project can be explained as a desktop application with separate layers for user interface, project data, ladder editing, I/O management, simulation logic, and packaging. This separation makes it easier to explain how a PLC software tool works.

Layer	Responsibility	Explanation
User interface	Shows menus, project tree, tabs, editor, simulator controls, and status panels.	This is what the user interacts with.
Project manager	Stores project name, ladder pages, I/O lists, and settings.	Keeps work organized.

Layer	Responsibility	Explanation
Ladder editor	Allows creating and modifying ladder rungs.	Represents control logic visually.
I/O manager	Handles digital input and output variables.	Connects logic to machine signals.
Simulation engine	Evaluates logic during simulation.	Shows how PLC scan logic affects outputs.
Packaging layer	Bundles the application as a portable EXE.	Makes deployment simple.

Module Explanation

Module	What it does	Why it matters
Home or project screen	Creates, opens, saves, or renames PLC learning projects.	Gives the user a starting point.
Project tree	Groups program files, I/O, settings, and documentation tabs.	Teaches structured engineering workflow.
Ladder canvas	Displays rungs and logic elements.	Makes PLC programming visual.
Instruction palette	Provides basic ladder elements such as contacts and coils.	Helps beginners build logic step by step.
I/O table	Lists inputs and outputs with names and current states.	Shows how real sensors and actuators are represented.
Simulation controls	Starts, stops, and resets the simulation.	Allows safe practice without equipment.
Status or log panel	Shows messages, errors, and actions.	Helps debugging and explanation.
About or copyright tab	Displays project ownership and identity.	Supports professional presentation.

PLC Concepts Explained By The Project

Concept	Meaning	How LADPORTAL helps
Digital input	A signal coming from a sensor, switch, or button.	Users can define and toggle inputs in simulation.
Digital output	A signal sent to a lamp, relay, motor starter, valve, or actuator.	Outputs show the result of logic evaluation.
Contact	A ladder element that checks a condition.	Users learn normally open and normally closed style thinking.
Coil	A ladder element that writes an output or internal bit.	Users see how logic drives output state.

Concept	Meaning	How LADPORTAL helps
Rung	One horizontal logic network.	The editor organizes logic in a readable ladder structure.
Scan cycle	PLC reads inputs, solves logic, then updates outputs repeatedly.	Simulation can demonstrate cause and effect.
Interlock	Logic that prevents unsafe or conflicting actions.	Users can build examples such as motor start/stop logic.

Working Principle

The working principle follows a simplified PLC workflow. The user creates a project, defines I/O points, builds ladder rungs, starts simulation, changes input states, and watches how the outputs react. This helps the learner understand the basic PLC scan cycle before working with real industrial software.

Step	What happens	Learning value
1	User opens the portable LADPORTAL application.	No complex setup is required for demonstration.
2	User creates or opens a PLC project.	Introduces project-based engineering workflow.
3	User defines input and output tags.	Connects software symbols to machine-style signals.
4	User builds ladder rungs with logic elements.	Practices PLC programming concepts.
5	User switches simulation on.	Begins safe software-only testing.
6	The simulator evaluates rung logic repeatedly.	Demonstrates the PLC scan cycle.
7	Outputs and monitoring states update on screen.	Shows real-time cause and effect.
8	User edits logic and tests again.	Builds troubleshooting confidence.

Example Demonstration Scenario

A simple demonstration can be a motor start and stop circuit. The user defines a start pushbutton, stop pushbutton, motor output, and holding contact. Then the ladder logic can be simulated to show how a momentary start signal can latch the motor until the stop input opens the circuit.

Signal	Example tag	Purpose
Input	I0.0 Start PB	Starts the simulated motor circuit.
Input	I0.1 Stop PB	Stops the simulated motor circuit.

Signal	Example tag	Purpose
Output	Q0.0 Motor	Represents the motor contactor output.
Internal logic	Motor seal-in branch	Keeps output active after start is released.
Simulation	Run mode	Lets the user test the behavior safely.

Testing And Validation

The supplied Windows package was validated statically in this workspace. The executable was not launched here. Final functional validation should be done on a Windows test machine using safe sample projects.

Test or check	How it was validated	Result or expected result
Package identification	Checked file type.	PE32 Windows GUI executable, NSIS self-extracting archive.
Version metadata	Read embedded version strings.	Product LADPORTAL, version 1.0.0.
Author metadata	Read copyright/version strings.	Bhushan Hiralal Bhole / Automation Club appears in metadata.
Package integrity	Calculated SHA-256 hash.	Stable hash recorded in package facts.
Windows launch test	Run on a Windows test machine.	Application should open without installation conflicts.
Project workflow test	Create, rename, save, and reopen a sample project.	Project data should remain consistent.
I/O simulation test	Toggle sample inputs and observe outputs.	Logic state should update correctly.
Error handling test	Try incomplete logic or missing tags.	Application should show clear feedback.

Challenges Faced And Solutions

Challenge	Solution used in the project
Making PLC logic understandable	Used a visual ladder-logic style workflow instead of only code.
Helping students without hardware	Built the project as a simulator and learning environment.
Keeping the interface beginner friendly	Focused on common project actions, I/O lists, and simulation controls.

Challenge	Solution used in the project
Connecting theory with practice	Used examples such as inputs, outputs, coils, rungs, and scan cycle.
Sharing the app easily	Packaged the project as a portable Windows executable.
Presenting ownership clearly	Included LADPORTAL and Automation Club identity in package metadata.

Results And Outcome

- A portable Windows build of LADPORTAL version 1.0.0 was produced.
- The project can be presented as a PLC ladder logic learning and simulation tool.
- The application identity is visible in file metadata as LADPORTAL by Automation Club desktop application.
- The project supports academic explanation, Automation Club demonstrations, and interview discussion.
- The project shows practical connection between automation theory and software development.

Learning Outcomes

This project helped me connect PLC programming fundamentals with application design. I learned that an automation tool needs more than logic blocks. It also needs a clear workflow, understandable naming, reliable simulation behavior, and packaging that makes the project easy to demonstrate.

- Improved understanding of PLC ladder logic and digital I/O representation.
- Practiced explaining the PLC scan cycle in a beginner-friendly way.
- Learned how project structure affects the usability of engineering software.
- Understood the importance of simulation before hardware testing.
- Practiced building a project that can be shown in automation interviews.

Limitations

- The project is a learning simulator, not a certified PLC programming suite.
- It should not be used to control real machines or safety functions.
- The portable EXE should be tested on Windows before final demonstration.
- Advanced PLC features such as fieldbus communication, online monitoring of real PLCs, and hardware download are outside the first version.
- Industrial behavior such as scan timing, memory areas, and vendor-specific instructions may be simplified.

Future Scope

Future improvement	Why it would improve the project
More ladder instructions	Would allow richer practice with timers, counters, comparisons, set/reset, and edge logic.
Structured project save format	Would make sharing and versioning student projects easier.
Built-in example library	Would help beginners learn motor control, conveyor, traffic light, and interlock examples.
Simulation trace view	Would show how each scan cycle changes inputs, memory bits, and outputs.
Export to PDF or image	Would help students submit ladder diagrams in reports.
PLC communication learning mode	Would explain Modbus, S7, or OPC UA concepts safely without controlling real hardware.
Windows installer signing	Would improve trust and professional distribution.

How I Can Explain This Project In An Interview

I built LADPORTAL as a portable PLC ladder logic simulator for automation learning. The idea was to create a simple desktop tool where a user can organize a PLC project, define inputs and outputs, build ladder logic, start simulation, and observe how the outputs change. It helped me connect PLC theory with software design. I would explain that it is not a replacement for professional PLC tools like TIA Portal, but it is a strong learning and portfolio project that shows my interest in industrial automation, PLC logic, and practical tool building.

Short CV Description

- Developed LADPORTAL, a portable Windows desktop application for PLC ladder logic learning and simulation.
- Designed a beginner-friendly PLC workflow covering project structure, digital I/O mapping, ladder logic, simulation controls, and monitoring.
- Packaged the application as LADPORTAL version 1.0.0 under Automation Club for demonstration and portfolio use.

Package Facts

Item	Observed value
Uploaded file	LADPORTAL-Portable-1.0.0.exe (3)
File size	98,897,123 bytes, approximately 95 MB
File type	PE32 executable GUI Intel 80386 for Microsoft Windows

Item	Observed value
Packaging type	Nullsoft Installer self-extracting archive with embedded 7z payload
PE sections	5 sections
Requested privilege	asInvoker, based on embedded manifest
File description	LADPORTAL by Automation Club desktop application
Product name	LADPORTAL
File version	1.0.0
Product version	1.0.0
Copyright metadata	Copyright 2026 Bhushan Hiralal Bhole / Automation Club
Embedded payload offset	50,504 bytes
Embedded payload size	98,846,619 bytes
SHA-256	17f07bd1cdd3c392392e5caf6f52a3b5412d53facb765c9184be922317207d98

Conclusion

LADPORTAL is a strong student automation project because it combines PLC ladder logic, simulation thinking, user interface design, and Windows application packaging. It can be presented as a learning tool for PLC fundamentals and as evidence of practical interest in industrial automation. The best way to present it is as an educational simulator and portfolio project, with future development focused on more ladder instructions, better reporting, richer examples, and verified Windows testing.